

PRACTICAL WEEK-5

TASK-1 : Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

```
GNU nano 8.7.1 producer_consumer.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <sys/time.h>

#define COUNT 100000

int main()
{
    int pipefd[2];
    pid_t pid;
    int data;
    struct timeval start, end;

    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        return 1;
    }

    gettimeofday(&start, NULL);

    pid = fork();

    if (pid == -1)
    {
        perror("fork");
        return 1;
    }

    if (pid > 0)
    {
        /* Parent - Producer */

        close(pipefd[0]);

        for (int i = 0; i < COUNT; i++)
        {
            data = i;
            write(pipefd[1], &data, sizeof(data));
        }

        close(pipefd[1]);

        wait(NULL);

        gettimeofday(&end, NULL);
    }
}
```

OUTPUT:

```
pavani@DESKTOP-380LEIN:~$ cd ossp
pavani@DESKTOP-380LEIN:~/ossp$ cd ~/week5
-bash: cd: /home/pavani/week5: No such file or directory
pavani@DESKTOP-380LEIN:~/ossp$ nano producer_consumer.c
pavani@DESKTOP-380LEIN:~/ossp$ gcc producer_consumer.c -o producer_consumer
pavani@DESKTOP-380LEIN:~/ossp$ ./producer_consumer
Consumer received 100000 integers.
Sum of received data: 4999950000

Communication completed.
Data transferred: 100000 integers
Time taken: 0.054568 seconds
Throughput: 7330303.47 bytes/second
pavani@DESKTOP-380LEIN:~/ossp$ █
```

REPORT: I implemented a producer-consumer communication system using an anonymous pipe, where the parent process generated data and the child process consumed it. The program successfully transferred 100,000 integers, and the communication time and throughput were measured. The program was successfully executed and tested in Ubuntu.

TASK-2:

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
GNU nano 8.7.1 pipe_grep.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];
    pid_t pid1, pid2;

    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        return 1;
    }

    // First child: ls -l
    pid1 = fork();

    if (pid1 == -1)
    {
        perror("fork");
        return 1;
    }

    if (pid1 == 0)
    {
        close(pipefd[0]);

        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[1]);

        execlp("ls", "ls", "-l", NULL);

        perror("execlp");
        exit(1);
    }

    // Second child: grep ".c"
    pid2 = fork();

    if (pid2 == -1)
    {
        perror("fork");
        return 1;
    }
}
```

OUTPUT:

```

pavani@DESKTOP-380LEIN:~/oss$ nano pipe_grep.c
pavani@DESKTOP-380LEIN:~/oss$ gcc pipe_grep.c -o pipe_grep
pavani@DESKTOP-380LEIN:~/oss$ ./pipe_grep
-rwxr-xr-x 1 pavani pavani 16632 Aug 20 05:51 builtin_dispatch
-rw-r--r-- 1 pavani pavani 1738 Aug 20 05:51 builtin_dispatch.c
-rw-r--r-- 1 pavani pavani 29 Aug 12 10:09 copy.txt
drwxr-xr-x 2 pavani pavani 4096 Aug 4 05:08 docs
-rwxr-xr-x 1 pavani pavani 16216 Aug 12 10:08 filecopy
-rw-r--r-- 1 pavani pavani 1032 Aug 12 10:08 filecopy.c
-rw-r--r-- 1 pavani pavani 443 Aug 4 05:36 fork_example.c
-rwxr-xr-x 1 pavani pavani 16040 Aug 4 05:34 fork_exec
-rw-r--r-- 1 pavani pavani 389 Aug 4 05:34 fork_exec.c
drwxr-xr-x 2 pavani pavani 4096 Aug 4 05:08 include
-rw-r--r-- 1 pavani pavani 1097 Aug 20 05:36 path_resolver.c
-rw-r--r-- 1 pavani pavani 1024 Aug 29 09:45 pipe_grep.c
-rwxr-xr-x 1 pavani pavani 16400 Aug 29 09:41 producer_consumer
-rw-r--r-- 1 pavani pavani 1713 Aug 29 09:41 producer_consumer.c
drwxr-xr-x 2 pavani pavani 4096 Aug 4 05:08 scripts
drwxr-xr-x 2 pavani pavani 4096 Aug 4 05:07 src
-rw-r--r-- 1 pavani pavani 3976 Aug 12 08:46 task2.c
-rw-r--r-- 1 pavani pavani 5425 Aug 12 08:50 task3.c
-rw-r--r-- 1 pavani pavani 9401 Aug 12 08:53 task4.c
-rw-r--r-- 1 pavani pavani 1527 Aug 12 09:09 task5.c
-rw-r--r-- 1 pavani pavani 2349 Aug 12 09:12 task6.c
-rw-r--r-- 1 pavani pavani 1041 Aug 12 09:14 task7.c
-rw-r--r-- 1 pavani pavani 1359 Aug 12 09:17 task8.c
-rw-r--r-- 1 pavani pavani 2343 Aug 20 05:41 variable_expansion.c
-rw-r--r-- 1 pavani pavani 1047 Aug 20 05:00 wait_waitpid.c
-rw-r--r-- 1 pavani pavani 957 Aug 20 05:32 waitpid_monitor.c
-rw-r--r-- 1 pavani pavani 670 Aug 20 05:02 zombie.c
-rw-r--r-- 1 pavani pavani 695 Aug 20 05:06 zombie_fixed.c
pavani@DESKTOP-380LEIN:~/oss$ ls -l | grep ".c"
-rwxr-xr-x 1 pavani pavani 16632 Aug 20 05:51 builtin_dispatch
-rw-r--r-- 1 pavani pavani 1738 Aug 20 05:51 builtin_dispatch.c
-rw-r--r-- 1 pavani pavani 29 Aug 12 10:09 copy.txt
drwxr-xr-x 2 pavani pavani 4096 Aug 4 05:08 docs
-rwxr-xr-x 1 pavani pavani 16216 Aug 12 10:08 filecopy
-rw-r--r-- 1 pavani pavani 1032 Aug 12 10:08 filecopy.c
-rw-r--r-- 1 pavani pavani 443 Aug 4 05:36 fork_example.c

```

```

pipe_grep
execve("./pipe_grep", ["/pipe_grep"], 0x7ffd30f9a868 /* 26 vars */) = 0
close(3) = 0
close(3) = 0
pipe2([3, 4], 0) = 0
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=0x782c4f563a10) = 718
strace: Process 718 attached
[pid 718] close(3 <unfinished ...>)
[pid 717] clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD <unfinished ...>)
[pid 718] <... close resumed> = 0
[pid 718] dup2(4, 1) = 1
[pid 718] close(4) = 0
strace: Process 719 attached
[pid 717] <... clone resumed>, child_tidptr=0x782c4f563a10) = 719
[pid 718] execve("/usr/local/sbin/ls", ["ls", "-l"], 0x7ffda94c4bc8 /* 26 vars */) = -1 ENOENT (No such file or directory)
[pid 717] close(3) = 0
[pid 718] execve("/usr/local/bin/ls", ["ls", "-l"], 0x7ffda94c4bc8 /* 26 vars */ <unfinished ...>)
[pid 717] close(4) = 0
[pid 718] <... execve resumed> = -1 ENOENT (No such file or directory)
[pid 717] wait4(718 <unfinished ...>)
[pid 718] execve("/usr/sbin/ls", ["ls", "-l"], 0x7ffda94c4bc8 /* 26 vars */ <unfinished ...>)
[pid 719] close(4 <unfinished ...>)
[pid 718] <... execve resumed> = -1 ENOENT (No such file or directory)
[pid 719] <... close resumed> = 0
[pid 718] execve("/usr/bin/ls", ["ls", "-l"], 0x7ffda94c4bc8 /* 26 vars */ <unfinished ...>)
[pid 719] dup2(3, 0) = 0
[pid 719] close(3) = 0
[pid 719] execve("/usr/local/sbin/grep", ["grep", ".c"], 0x7ffda94c4bc8 /* 26 vars */) = -1 ENOENT (No such file or directory)
[pid 719] execve("/usr/local/bin/grep", ["grep", ".c"], 0x7ffda94c4bc8 /* 26 vars */) = -1 ENOENT (No such file or directory)
[pid 718] <... execve resumed> = 0
[pid 719] execve("/usr/sbin/grep", ["grep", ".c"], 0x7ffda94c4bc8 /* 26 vars */) = -1 ENOENT (No such file or directory)
[pid 719] execve("/usr/bin/grep", ["grep", ".c"], 0x7ffda94c4bc8 /* 26 vars */) = 0
[pid 718] close(3) = 0
[pid 719] close(3) = 0
[pid 718] close(3) = 0
[pid 719] close(3) = 0
[pid 718] close(3) = 0

```

REPORT: I developed a program to execute the equivalent of `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls. The output of `ls -l` was passed through the pipe to `grep` for filtering. The program was successfully compiled, executed, and verified in Ubuntu.